

PARCELING CANNOT REDUCE FACTOR INDETERMINACY IN FACTOR ANALYSIS: A RESEARCH NOTE

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Parceling—using composites of observed variables as indicators for a common factor—strengthens loadings, but reduces the number of indicators. Factor indeterminacy is reduced when there are many observed variables per factor, and when loadings and factor correlations are strong. It is proven that parceling cannot reduce factor indeterminacy. In special cases where the ratio of loading to residual variance is the same for all items included in each parcel, factor indeterminacy is unaffected by parceling. Otherwise, parceling worsens factor indeterminacy. While factor indeterminacy does not affect the parameter estimates, standard errors, or fit indices associated with a factor model, it does create uncertainty, which endangers valid inference.

Key words: factor analysis, parceling, factor indeterminacy, uncertainty.

Introduction

Parceling is the practice of using the sums or means of observed variables (or items), rather than the observed variables themselves, as indicators in a factor analysis model. Parceling is credited with averaging out item-level discrepancies between a factor model and the data, improving normality, and reducing the number of parameters estimated, thereby potentially improving model fit without necessarily biasing results (e.g., Bandalos, 2002; Plummer, 2000; Sass & Smith, 2006). Reviews indicate that, across various journals, between 17 and 74% of all papers with confirmatory factor analysis or structural equation modeling applications also employ parceling (Bandalos & Finney, 2001; Hall, Snell & Foust, 1999; Plummer, 2000; Williams & O’Boyle, 2008). In fact, parceling is currently considered the “prevailing approach for including scales with many items in factor analysis and structural equation modeling” (Marsh, Lüdtke, Nagengast, Morin, & Von Davier, 2013, p. 258; see also Yang, Nay, & Hoyle, 2010).

Factor analysis comes with the phenomenon of factor indeterminacy. First identified by Wilson (1928), factor indeterminacy means that a common factor cannot be faithfully reproduced as one unique best function of the observed variables in the factor model. While there have been

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substantial arguments over the implications of factor indeterminacy (see, for example, Maraun, 1996, with replies and rejoinders), the mathematics of the problem are well established (Guttman, 1955; Schönemann & Wang, 1972). Recent research argues that factor indeterminacy threatens the validity of measurement (Rigdon, Becker, & Sarstedt, 2019). While factor indeterminacy does not directly affect fit indices, parameter estimates or standard errors *within* a particular factor model (Schönemann & Haagen, 1987), it induces a band of uncertainty around the correlation between a common factor and any variable outside the model (Schönemann & Steiger, 1978; Steiger, 1979), which may imperil the validity of generalizations beyond individual studies.

While numerous studies have discussed the concepts of parceling and factor indeterminacy in isolation, none have addressed the interplay between parceling and factor indeterminacy. Parceling can increase correlations among indicators and thus increase factor loadings, which will tend to reduce factor indeterminacy. At the same time, however, parceling reduces the number of indicators in a factor model, which exacerbates indeterminacy (Rigdon et al., 2019). This research note addresses the overall implications of parceling for factor indeterminacy. We show analytically that parceling cannot reduce factor indeterminacy. At best, it leaves factor indeterminacy unchanged, but more likely, current parceling practice will worsen factor indeterminacy.

Review of Concepts

Common factor model In a common factor model, P observed variables $\mathbf{Y}$ (y_p , $p = 1$ to P) are related to K common factors $\mathbf{F}$ (f_k , $k = 1$ to K) and P unique factors $\mathbf{U}$ (u_p , $p = 1$ to P) according to:

$$\mathbf{Y} = \mathbf{\Lambda}\mathbf{F} + \mathbf{U} \quad (1)$$

assuming 0 vectors for the means of $\mathbf{Y}$, $\mathbf{F}$ and $\mathbf{U}$. Here $\mathbf{\Lambda}$ is a $P \times K$ matrix of loadings. Assume that the indicators $\mathbf{Y}$ are congeneric, each observed variable loading on a single common factor, with no residual covariance. Crucially, assume that the unique factors $\mathbf{U}$ are orthogonal to the common factors $\mathbf{F}$. Then Eq. 1 implies a covariance matrix for $\mathbf{Y}$ of the form:

$$\mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Phi}\mathbf{\Lambda}' + \mathbf{\Theta} \quad (2)$$

where $\mathbf{\Sigma}$, $\mathbf{\Phi}$ and $\mathbf{\Theta}$ are covariance matrices for $\mathbf{Y}$, $\mathbf{F}$ and $\mathbf{U}$, respectively.

Factor indeterminacy Augment Eq. 1 with an explicit unit loading for each observed variable in $\mathbf{Y}$ on its associated specific factor in $\mathbf{U}$:

$$\mathbf{Y} = \mathbf{\Lambda}\mathbf{F} + \mathbf{I}\mathbf{U} \quad (3)$$

where $\mathbf{I}$ is an identity matrix of dimension P . Then Eq. 2 can be helpfully expressed in terms of partitioned matrices:

$$\begin{aligned} \mathbf{\Sigma} &= [\mathbf{\Lambda} \mathbf{I}] \begin{bmatrix} \mathbf{\Phi} & \mathbf{0}' \\ \mathbf{0} & \mathbf{\Theta} \end{bmatrix} [\mathbf{\Lambda} \mathbf{I}]' \\ \mathbf{\Sigma} &= [\mathbf{\Lambda} \mathbf{I}] \mathbf{W} [\mathbf{\Lambda} \mathbf{I}]' \end{aligned} \quad (4)$$

where $\mathbf{0}$ is a matrix of zeroes of dimension $P \times K$. Equation 4 illustrates the core of the factor indeterminacy problem. On the left side, the covariance matrix of the P observed variables will have rank at most P . The rank of the covariance matrix indicates “the maximum number of linearly independent rows (columns)” (Mulaik, 2010, p. 47), and hence the maximum number

of independent dimensions of information in the data. But the assumed orthogonality of $\mathbf{F}$ and $\mathbf{U}$ means that the matrix $\mathbf{W}$ may have rank up to $P + K$, implying that the model has more dimensions of information than the data (Schönemann & Steiger, 1978, pp. 287–288).

When $\text{rank}(\mathbf{W}) > \text{rank}(\mathbf{\Sigma})$, the common and unique factors are indeterminate, meaning that each variable in $\mathbf{F}$ and $\mathbf{U}$ includes an arbitrary component. Solving Eq. 4 for $\mathbf{F}$ and $\mathbf{U}$ yields (Guttman, 1955):

$$\begin{aligned}\mathbf{F} &= \mathbf{\Phi} \mathbf{\Lambda}' \mathbf{\Sigma}^{-1} \mathbf{Y} + \mathbf{BZ} \\ \mathbf{U} &= \mathbf{\Theta} \mathbf{\Sigma}^{-1} \mathbf{Y} - \mathbf{\Lambda} \mathbf{BZ}\end{aligned}\quad (5)$$

Here, $\mathbf{Z}$ is a set of K orthonormal variables z_k , $k = 1$ to K , one per common factor (Guttman, 1955, p. 72). There are an infinite number of sets of numerical values for each z_k . Subject to the orthonormality constraint, every valid choice for $\mathbf{Z}$ will produce $\mathbf{F}$ and $\mathbf{U}$ that exactly reproduce the factor model. Failing to include $\mathbf{Z}$ in the construction of common and unique factors induces bias in relations among those variables (Lastovicka & Thamodaran, 1991; Skrondal & Laake, 2001).

The $K \times P$ parameter matrix $\mathbf{B}$ in Eq. 5 is such that:

$$\mathbf{B} \mathbf{B}' = \mathbf{\Phi} - \mathbf{\Phi} \mathbf{\Lambda}' \mathbf{\Sigma}^{-1} \mathbf{\Lambda} \mathbf{\Phi} \quad (6)$$

In Eq. 6, the second term on the right-hand side represents the portion of the covariance matrix of the factors which is accounted for by the data, given the model. When standardized, the k^{th} diagonal element of that term is equal to $\rho_{f_k, Y}^2$, the squared correlation of the k^{th} common factor, conditional on the model's observed variables (Mulaik, 2010). As a whole, then, the right-hand side of Eq. 6 represents the unexplained portion of the factor covariance matrix.

Guttman (1955) defined an index of factor indeterminacy as

$$\rho_{\min} = 2\rho_{f_k, Y}^2 - 1 \quad (7)$$

with $\rho_{\min}$ representing the minimum correlation between alternate sets of factor scores for the same common factor, model and data. This correlation need not be large or even positive. Rank orderings of cases by factor score can vary across these different sets of factor scores, giving pause regarding the use of such scores for a range of purposes.

Parcels In the most general terms, parceling involves replacing the P observed variable indicators $\mathbf{Y}$ with $Q < P$ parceled indicators, using a weight matrix $\mathbf{V}_{P \times Q}$, yielding a new set of indicators $\mathbf{V}'\mathbf{Y}$. Then Eqs. 1 and 2 can be re-expressed as:

$$\mathbf{V}'\mathbf{Y} = \mathbf{V}'\mathbf{\Lambda}\mathbf{F} + \mathbf{V}'\mathbf{U} \quad (8)$$

$$\mathbf{\Sigma}_v = \mathbf{V}'\mathbf{\Lambda}\mathbf{\Phi}\mathbf{\Lambda}'\mathbf{V} + \mathbf{V}'\mathbf{\Theta}\mathbf{V} \quad (9)$$

where $\mathbf{\Sigma}_v$ indicates the new covariance matrix of the parcels.

The parcel weight matrix $\mathbf{V}$ could be very general, in principal. In practice, parceling almost always involves creating sums or averages of observed variables (e.g., Bandalos, 2002; Bandalos & Finney, 2001). This implies equal weighting of the observed variables, most commonly with a unit weight. In addition, each row in $\mathbf{V}$ (i.e., each observed variable) is usually assigned to only one column (i.e., one parcel) via a non-zero weight.

The question of which observed variables to include in which parcel has been the subject of considerable debate in the literature, giving rise to numerous approaches to parceling

(for an overview, see, for example, Landis, Beal, & Tesluk, 2000; Little, Cunningham, Shahar, & Widaman, 2002; and Rhemtulla, 2016). Recent research distinguishes between two general approaches (Marsh et al., 2013): *homogeneous parceling* (also termed *isolated* parceling; Bandalos, 2008; Rhemtulla, 2016) and *distributive parceling*. In homogeneous parceling, closely related observed variables that share, for example, common characteristics such as item wordings, systematic variation, or large correlations, are placed in the same parcel (Marsh et al., 2013; Rhemtulla, 2016). While the observed variables within a parcel are all homogeneous, the different parcels are maximally different from each other. In contrast, in distributive parceling, closely related observed variables that share a source of systematic variation are systematically or randomly distributed across different parcels, creating parcels that are similar to each other (Rhemtulla, 2016). Both strategies have their advantages and disadvantages and in general no approach is widely preferred, despite considerable concerns expressed about distributive parceling strategies (e.g., Marsh et al., 2013).

Parceling and Factor Indeterminacy

Factor covariances play a role in the degree of factor indeterminacy, as can be seen from Eqs. 5 and 6. However, parceling is known to not change structural parameters, including factor covariances, if the item-level model is correct (Sass & Smith, 2006), so it seems efficient to focus on models having a single common factor. Assume that a congeneric factor model is correct at the item level, so that parceling will not change the nature of the common factors.

Assume also that the single common factor is standardized to zero mean and unit variance. In the case of a single common factor, this standardization means that Eq. 2 becomes:

$$\Sigma = \lambda\lambda' + \Theta \quad (10)$$

and $\rho_{f,Y}^2$, or the single element in $\mathbf{BB}'$ from Eq. 6, becomes:

$$\rho_{f,Y}^2 = \lambda' \Sigma^{-1} \lambda \quad (11)$$

With Θ diagonal and nonsingular, following Gentle (2017, p. 120, Equation 3.185):

$$\Sigma^{-1} = (\lambda\lambda' + \Theta)^{-1} = \Theta^{-1} - \frac{\Theta^{-1}\lambda\lambda'\Theta^{-1}}{1 + \lambda'\Theta^{-1}\lambda} \quad (12)$$

Equation 11 can then be re-expressed as:

$$\rho_{f,Y}^2 = \lambda' \left(\Theta^{-1} - \frac{\Theta^{-1}\lambda\lambda'\Theta^{-1}}{1 + \lambda'\Theta^{-1}\lambda} \right) \lambda \quad (13)$$

Distributing the terms:

$$\rho_{f,Y}^2 = \lambda' \Theta^{-1} \lambda - \frac{\lambda' \Theta^{-1} \lambda \lambda' \Theta^{-1} \lambda}{1 + \lambda' \Theta^{-1} \lambda} \quad (14)$$

Let $\mathbf{A} = \lambda' \Theta^{-1} \lambda$. Then:

$$\rho_{f,Y}^2 = \mathbf{A} - \frac{\mathbf{A}^2}{1 + \mathbf{A}} = 1 - \frac{1}{1 + \mathbf{A}}$$

Therefore:

$$\rho_{f,Y}^2 = 1 - \frac{1}{1 + \lambda' \Theta^{-1} \lambda} \quad (15)$$

This implies that $\rho_{f,Y}^2$ (the complement to factor indeterminacy) monotonically increases with $\lambda' \Theta^{-1} \lambda$, that is, with higher loadings and smaller error variances.

From Eqs. 11 and 9, the same coefficient for parcels will be:¹

$$\rho_{f,Y_v}^2 = 1 - \frac{1}{1 + \lambda' \mathbf{V}(\mathbf{V}' \Theta \mathbf{V})^{-1} \mathbf{V}' \lambda} \quad (16)$$

The impact of parceling on factor indeterminacy amounts to the difference between $\lambda' \Theta^{-1} \lambda$ and $\lambda' \mathbf{V}(\mathbf{V}' \Theta \mathbf{V})^{-1} \mathbf{V}' \lambda$.

Define a new matrix

$$\mathbf{X} = \Theta^{1/2} \mathbf{V} \quad (17)$$

Then the parceled equivalent of $\lambda' \Theta^{-1} \lambda$ can be expressed as:

$$\lambda' \Theta^{-1/2} \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \Theta^{-1/2} \lambda \quad (18)$$

and equality can be established via:

$$\lambda' \Theta^{-1} \lambda = \lambda' \Theta^{-1/2} \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \Theta^{-1/2} \lambda + \lambda' \Theta^{-1/2} [\mathbf{I} - \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'] \Theta^{-1/2} \lambda \quad (19)$$

The term $\mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'$ is familiar from regression. It is known to be an idempotent matrix, as is $\mathbf{I} - \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'$. Idempotent matrices are always positive semi-definite (Gentle 2017, p. 353), meaning that for any non-zero column vector $\mathbf{z}$ the product $\mathbf{z}'(\mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}')\mathbf{z}$ cannot be less than zero. Thus, both products on the right-hand side of Eq. 19 must be nonnegative, with the left-hand side being equal to or greater than either term on the right. Hence:

$$\lambda' \Theta^{-1} \lambda \geq \lambda' \Theta^{-1/2} \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \Theta^{-1/2} \lambda = \lambda' \mathbf{V}(\mathbf{V}' \Theta \mathbf{V})^{-1} \mathbf{V}' \lambda \quad (20)$$

and

$$\rho_{f,Y}^2 \geq \rho_{f,Y_v}^2 \quad (21)$$

QED. Factor indeterminacy in the parceled model can never be less than factor indeterminacy in the original model.

However, parceling does not always exacerbate factor indeterminacy. The analogy to regression, inspired by the term $\mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'$, suggests general conditions under which parceling will have no effect on indeterminacy. If $\Theta^{-1/2} \lambda$ were the dependent variable regressed on the columns of $\mathbf{X}$, then the two terms on the right-hand side of Eq. 19 would represent the explained part and the unexplained part of the variance of $\Theta^{-1/2} \lambda$, respectively. If the parceling weight matrix, $\mathbf{V}$, were chosen so that there is no residual from this regression, then the second term of Eq. 19 would vanish and $\rho_{f,Y}^2 = \rho_{f,Y_v}^2$. This would require:

$$\Theta^{-1/2} \lambda = \mathbf{X} \beta = \Theta^{1/2} \mathbf{V} \beta \quad (22)$$

¹ An anonymous reviewer suggested this substantially generalized proof, the analogy to regression, and the application of that analogy to identifying circumstances where parceling does not worsen factor indeterminacy. We very much thank the reviewer for these remarkable insights.

or

$$\Theta^{-1}\lambda = \mathbf{V}\beta \quad (23)$$

for some β , which will be true if the parceling weights in $\mathbf{V}$ are proportional to the original item loadings divided by the original residual variances.

This can happen in a variety of circumstances. Suppose that the original observed variables are standardized with equal loadings, and hence with equal residual variances. Then the ratio of loading to residual variance will also be equal across observed variables. In such a case, the equal weights typically employed in parceling will meet the condition described above, and factor indeterminacy will be identical, at the population level, for both the original model and the parceled model.

The same would be true if the original observed variables have different loadings, but the researcher constructs parcels by combining items whose original loadings (and residual variances) are identical. Such a situation represents an ideal example of *homogeneous parceling*, and in such a situation, using equal weights would not worsen factor indeterminacy. On the contrary, in *distributive parceling*, using equal parceling weights would always worsen indeterminacy. However, using differential parceling weights—weights proportional to each item's loading divided by its residual variance (Eq. 23)—offers a means for using distributive parceling without making factor indeterminacy worse.

Discussion

This research note demonstrates that parceling in factor analysis at best leaves factor indeterminacy unchanged. In homogeneous parceling, which involves forming parcels out of closely related items (Marsh et al., 2013), using equal parceling weights will not worsen factor indeterminacy if items in a parcel have the same loading and residual variance in the original model. Nevertheless, as it is unlikely that the loadings and residual variances within a parcel will ever be perfectly equal, using equal parceling weights means that factor indeterminacy is likely to worsen. Distributive parceling, where closely related observed variables are systematically or randomly distributed across different parcels, always worsens indeterminacy when using equal parceling weights. Considering that “many approaches to forming parcels (e.g., random assignment) are implicit (or de facto) distributive strategies even if this is not the explicit intent of the applied researcher” (Marsh et al., 2013, p. 260), this finding may be concerning.

Table 1 illustrates the impact of distributive parceling on factor indeterminacy, as indexed by Guttman's $\rho_{\min}$ (the minimum correlation between alternate sets of factor scores; Eq. 7). The original and parceled models produce identical factor indeterminacy values when the original loadings are equal. The greater the differences in original loadings, however, the lower the value of Guttman's $\rho_{\min}$ and the worse the indeterminacy in the parceled model.

Table 2 presents a more general analysis of a large random enumeration of a single factor model with 12 (Panel A) and 6 (Panel B) observed variables with different (random) loadings parceled with three different strategies: (1) distributive parceling with unequal loadings and proportional parceling weights, as suggested by Eq. 23, (2) homogeneous parceling combining maximally correlated observed variables into the same parcel, with unit parceling weights (Landis et al., 2000; as implemented in the R *psych* package; Revelle, 2018), and (3) distributive parceling using random assignment to parcels, with unit parceling weights. The results confirm that strategy (1) leaves indeterminacy unchanged, while strategies (2) and (3) worsen indeterminacy (i.e., they reduce $\rho_{\min}$). In addition, distributive parceling using random assignment and unit weights worsens indeterminacy more than does homogeneous parceling. From our analysis, we would expect that more extreme forms of distributive parceling, which explicitly construct parcels from the most divergent observed variables, will increase factor indeterminacy even more. The examples

TABLE 1.
Effect of distributive parceling on factor indeterminacy, unequal loadings.

Original loadings	Average difference	Original average loading	Loading for each parcel	Factor indeterminacy in the original model ($\rho_{\min}$)	Factor indeterminacy in the parceled model ($\rho_{\min}$)	Factor indeterminacy difference
6 Indicators (3 high loadings & 3 low loadings), 3 Parcels with 2 items each (one high & one low loading)						
.90 / .90	.00	.90	.946	.925	.925	0
.90 / .80	.10	.85	.917	.895	.881	.015
.90 / .70	.20	.80	.886	.880	.833	.047
.90 / .60	.30	.75	.855	.871	.781	.090
.90 / .50	.40	.70	.822	.865	.724	.140
.90 / .40	.50	.65	.788	.861	.662	.198
.90 / .30	.60	.60	.753	.858	.594	.264
.90 / .20	.70	.55	.716	.856	.519	.337
6 Indicators (3 high loadings & 3 low loadings), 3 Parcels with 2 items each (one high & one low loading)						
.50 / .50	.00	.50	.632	.333	.333	0
.60 / .40	.20	.50	.635	.386	.339	.047
.70 / .30	.40	.50	.643	.521	.357	.164
.80 / .20	.60	.50	.657	.690	.389	.301
.90 / .10	.80	.50	.677	.855	.435	.420
9 Indicators (3 high loadings, 3 medium loadings, 3 low loadings), 3 Parcels with 3 items each (one high, medium, & low loading)						
.90 / .80 / .70	.10	.80	.919	.909	.884	.025
.90 / .70 / .50	.20	.70	.868	.887	.802	.084
.80 / .70 / .60	.10	.70	.863	.817	.795	.021
.80 / .60 / .40	.20	.60	.799	.767	.682	.086

suggest that the impact of parceling can be sizeable, especially when equal weights are used to build parcels out of variables with substantially different loading / residual variance ratios.

Despite these concerns, researchers may well wonder whether they need to be concerned about factor indeterminacy at all, given that the factor analysis literature hardly considers the phenomenon worth quantifying. Factor indeterminacy does not directly affect parameter estimates or standard errors within a particular model, or model fit indices (Schönemann & Haagen, 1987). Moreover, researchers who view factor indeterminacy as a problem which only affects factor scores (the rendering of a model's common factors as specific functions of the model's observed variables) will note that most factor analysis exercises can be conducted without generating factor scores at all. In addition, if factor analysis is just a matter of data reduction, then factor indeterminacy may be viewed as “more an inconvenience than a limitation” (Hoyle, 2014, p. 4).

Viewed from another perspective, however, factor indeterminacy is a crucial threat to valid inference in factor analysis and structural equation modeling (Rigdon et al., 2019). The core of this argument has to do with inferences connecting the common factors within a model to variables outside that model. Factor indeterminacy creates a zone of uncertainty regarding the correlation between a common factor and any variable outside the factor model (Steiger, 1979). The correlation between a common factor and outside variable cannot be known with any greater precision than that defined in part by the factor indeterminacy associated with the common factor—the greater the indeterminacy, the wider the range of uncertainty.

This uncertainty endangers inference in multiple ways. Factor indeterminacy limits the ability of researchers to generalize across studies. When building networks of structural relations

TABLE 2.
Effect of different parceling strategies on factor indeterminacy.

	Original	Proportional weights	psych package	Random
<i>Panel A</i>				
Mean ($\rho_{\min}$)	.824	.824	.815	.768
SD ($\rho_{\min}$)	.051	.051	.055	.060
Min ($\rho_{\min}$)	.530	.530	.524	.499
Max ($\rho_{\min}$)	.924	.924	.923	.920
Median ($\rho_{\min}$)	.834	.834	.825	.775
<i>Panel B</i>				
Mean ($\rho_{\min}$)	.659	.659	.639	.590
SD ($\rho_{\min}$)	.130	.130	.135	.135
Min ($\rho_{\min}$)	−.044	−.044	−.045	−.056
Max ($\rho_{\min}$)	.898	.898	.894	.890
Median ($\rho_{\min}$)	.688	.688	.666	.607

Results are from 10,000 draws of loadings from a uniform distribution between .30 and .90 for 12 (Panel A) or 6 (Panel B) observed variables on one congeneric factor. Observed variables were combined into 4 parcels of 3 variables each (Panel A), or 3 parcels of 2 variables each (Panel B), by means of three different strategies: (1) using proportional weights as suggested in Eq. 23; (2) using the psych package in *R* which parcels items by their largest correlations, with unit weights; (3) using random assignment, with unit weights.

from models which only partially overlap, higher factor indeterminacy will mean wider gaps in knowledge. Replications of the “same” study with new populations will be more difficult if there is greater uncertainty surrounding each common factor.

There is a more fundamental problem if factor analysis and structural equation modeling are viewed from a scientific realist perspective (Borsboom, 2005; Haig & Evers, 2016; Psillos, 1999). From this viewpoint, the common factor in the statistical model is a representation of an actual trait or attribute—a real entity with its own existence independent of data. The validity of the representation—and the validity of inferences based on that representation—depends on the strength of the correlation between the common factor in the model and the entity being represented. Since the entity being represented is itself outside the factor model, the uncertainty due to factor indeterminacy also applies to the correlation between the common factor representation and the entity being represented. If that uncertainty is large, then inferences which take the common factor as proxy for the unobserved entity are dubious (Rigdon et al., 2019).

These results might be taken to suggest a superior parceling strategy involving the use of unequal weights proportional to the ratio of loading to residual variance for each original observed variable. Of course, actually obtaining the estimated loadings and residual variances involves estimating the model at the original item level, which likely obviates the need to estimate a parceled model (e.g., Marsh et al., 2013; Marsh, Hau, Balla, & Grayson, 1998). The larger point here is simply that there is “no free lunch”—parceling involves a sacrifice of information and leaves the researcher with more uncertainty, assuming that the model is correct for the original items.

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References

- Bandalos, D. L. (2002). The effects of item parceling on goodness-of-fit and parameter estimate bias in structural equation modeling. *Structural Equation Modeling*, 9(1), 78–102. https://doi.org/10.1207/S15328007SEM0901_5.

- Bandalos, D. L. (2008). Is parceling really necessary? A comparison of results from item parceling and categorical variable methodology. *Structural Equation Modeling*, 15(2), 211–240. <https://doi.org/10.1080/10705510801922340>.
- Bandalos, D. L., & Finney, S. J. (2001). Item parceling issues in structural equation modeling. In G. A. Marcoulides & R. E. Schumacker (Eds.), *New developments and techniques in structural equation modeling* (pp. 269–296). Mahwah: LEA.
- Borsboom, D. (2005). *Measuring the mind: Conceptual issues in contemporary psychometrics*. Cambridge: Cambridge University.
- Gentle, J. E. (2017). *Matrix algebra* (2nd ed.). Cham: Springer.
- Guttman, L. (1955). The determinacy of factor score matrices with implications for five other basic problems of common-factor theory. *British Journal of Statistical Psychology*, 8(2), 65–81. <https://doi.org/10.1111/j.2044-8317.1955.tb00321.x>.
- Haig, B. D., & Evers, C. W. (2016). *Realist inquiry in social science*. London: Sage.
- Hall, R. J., Snell, A. F., & Foust, M. S. (1999). Item parceling strategies in SEM: Investigating the subtle effects of unmodeled secondary constructs. *Organizational Research Methods*, 2(3), 233–256. <https://doi.org/10.1177/109442819923002>.
- Hoyle, R. H. (Ed.). (2014). *Handbook of structural equation modeling*. New York: Guilford Press.
- Landis, R. S., Beal, D. J., & Tesluk, P. E. (2000). A comparison of approaches to forming composite measures in structural equation models. *Organizational Research Methods*, 3(2), 186–207. <https://doi.org/10.1177/109442810032003>.
- Lastovicka, J. L., & Thamodaran, K. (1991). Common factor score estimates in multiple regression problems. *Journal of Marketing Research*, 28(1), 105–112. <https://doi.org/10.1177/002224379102800109>.
- Little, T. D., Cunningham, W. A., Shahar, G., & Widaman, K. F. (2002). To parcel or not to parcel: Exploring the question, weighing the merits. *Structural Equation Modeling*, 9(2), 151–173. https://doi.org/10.1207/S15328007SEM0902_1.
- Maraun, M. D. (1996). Metaphor taken as math: Indeterminacy in the factor analysis model. *Multivariate Behavioral Research*, 31(4), 517–538. https://doi.org/10.1207/s15327906mbr3104_6.
- Marsh, H. W., Lüdtke, O., Nagengast, B., Morin, A. J., & Von Davier, M. (2013). Why item parcels are (almost) never appropriate: Two wrongs do not make a right—Camouflaging misspecification with item parcels in CFA models. *Psychological Methods*, 18(3), 257–284. <https://doi.org/10.1037/a0032773>.
- Marsh, H. W., Hau, K.-T., Balla, J. R., & Grayson, D. (1998). Is more ever too much? The number of indicators per factor in confirmatory factor analysis. *Multivariate Behavioral Research*, 33(2), 181–220. https://doi.org/10.1207/s15327906mbr3302_1.
- Mulaik, S. A. (2010). *Foundations of factor analysis* (2nd ed.). Boca Raton: Chapman & Hall.
- Plummer, B. (2000). *To parcel or not to parcel: The effects of item parceling in confirmatory factor analysis*. Providence: Dissertation, The University of Rhode Island.
- Psillos, S. (1999). *Scientific realism: How science tracks truth*. Abingdon: Routledge.
- Revelle, W. (2018). psych: Procedures for personality and psychological research. Evanston: Northwestern University. <https://CRAN.R-project.org/package=psychVersion=1.8.12>. Accessed 27 Jan 2019.
- Rhemtulla, M. (2016). Population performance of SEM parceling strategies under measurement and structural model misspecification. *Psychological Methods*, 21(3), 348–368. <https://doi.org/10.1037/met0000072>.
- Rigdon, E. E., Becker, J. M., & Sarstedt, M. (2019). Factor indeterminacy as metrological uncertainty: Implications for advancing psychological measurement. *Multivariate Behavioral Research*, 54(3), 429–443. <https://doi.org/10.1080/00273171.2018.1535420>.
- Sass, D. A., & Smith, D. L. (2006). The effects of parceling unidimensional scales on structural parameter estimates in structural equation modeling. *Structural Equation Modeling*, 13(4), 566–586. https://doi.org/10.1207/s15328007sem1304_4.
- Schönemann, P. H., & Haagen, K. (1987). On the use of factor scores for prediction. *Biometrical Journal*, 29(7), 835–847. <https://doi.org/10.1002/bimj.4710290712>.
- Schönemann, P. H., & Steiger, J. H. (1978). On the validity of indeterminate factor scores. *Bulletin of the Psychonomic Society*, 12(4), 287–290. <https://doi.org/10.3758/BF03329685>.
- Schönemann, P. H., & Wang, M.-M. (1972). Some new results on factor indeterminacy. *Psychometrika*, 37(1), 61–91. <https://doi.org/10.1007/BF02291413>.
- Skrondal, A., & Laake, P. (2001). Regression among factor scores. *Psychometrika*, 66(4), 563–575. <https://doi.org/10.1007/BF02296196>.
- Steiger, J. H. (1979). The relationship between external variables and common factors. *Psychometrika*, 44(1), 93–97. <https://doi.org/10.1007/BF02293788>.
- Williams, L. J., & O'Boyle, E. H., Jr. (2008). Measurement models for linking latent variables and indicators: A review of human resource management research using parcels. *Human Resource Management Review*, 18, 233–242. <https://doi.org/10.1016/j.hrmr.2008.07.002>.
- Wilson, E. B. (1928). The abilities of man: Their nature and measurement. *Science*, 67(1731), 244–248.
- Yang, C., Nay, S., & Hoyle, R. H. (2010). Three approaches to using lengthy ordinal scales in structural equation models: Parceling, latent scoring, and shortening scales. *Applied Psychological Measurement*, 34(2), 122–142. <https://doi.org/10.1177/0146621609338592>.